Deep Learning & GenAI — Mock Paper with Solutions

Deep Learning & GenAI — Mock Final Exam (with Solutions)

> ***Format mirrors the 2025-11-30 IIT Bombay ePGD model paper.*** *Total:* ***50 marks****, suggested duration* ***3 hours****. Try Section 1 in ~30 min, Sections 2–11 in ~150 min. Solutions are at the end of each question in italic indented form — cover them while solving.*

Section 1 — Short answers / MCQs (20 × 1 = 20 marks)

For each, give a short answer or pick the option with a ≤3-line justification.

1.1 True/False: word2vec's skip-gram model with negative sampling pushes the focus-word vector *toward* the true context word and *away from* the sampled noise words. **(1)**

> *True. The skip-gram NEG objective* `-log σ(u_o·v_c) - Σ log σ(-u_n·v_c)` *maximises similarity to* `u_o` *and minimises similarity to each* `u_n`.

1.2 Which is the primary reason a single self-attention head is insufficient and we use Multi-Head Attention? A. Softmax saturates with large `d_k`. B. Softmax forces the head to be peaky, attending to one dominant relation. C. Single heads cannot be parallelised on a GPU. D. Single heads require quadratic memory. **(1)**

> ***B.*** *A softmax-attention head naturally concentrates mass. Multiple heads operate in different subspaces so subject, object, coreference, etc. can be attended simultaneously.*

1.3 True/False: In the original Transformer, positional encodings are *added* to the token embeddings, not concatenated. **(1)**

> *True. Same dimensionality* `d_model`*; added so that downstream linear projections can learn to separate the two contributions if needed.*

1.4 Which model's pre-training objective is **span corruption** with sentinel tokens? A. BERT B. GPT C. T5 D. BART **(1)**

> ***C — T5.*** *BERT uses token-level MLM; BART uses multi-noise denoising; T5 specifically replaces contiguous spans with sentinel tokens.*

1.5 True/False: A larger beam size in beam search generally yields a sequence of *higher* total probability under the model. **(1)**

> *True. Beam search keeps the top-B partial hypotheses each step; widening the beam never excludes hypotheses considered by a smaller beam, so the returned sequence is at least as probable.*

1.6 Among the RAG variants, in which is the final answer constructed by marginalising **per token** over multiple retrieved passages? A. REALM B. RAG-Sequence C. RAG-Token D. FiD **(1)**

> ***C — RAG-Token.*** *Each output token can mix evidence from different passages. RAG-Sequence marginalises per-output (single doc per output); FiD fuses inside the decoder but is not formally a marginalisation.*

1.7 True/False: BERT can be used directly (without modification) for left-to-right text generation comparable to GPT. **(1)**

> *False. BERT is trained with bidirectional MLM; it has no causal mask and is not optimised for next-token prediction. Generation requires architectural adaptation (e.g. UniLM) or a different model class.*

1.8 Which decoding strategy produces the smallest set of tokens whose cumulative probability exceeds `p`, sampling from within? A. Greedy B. Top-k C. Top-p (nucleus) D. Beam **(1)**

> ***C — Top-p (nucleus).*** *Top-k uses a fixed `k`; nucleus uses an adaptive set determined by cumulative probability.*

1.9 In an LSTM, when the forget gate `f_t ≈ 1` and input gate `i_t ≈ 0`, what happens to the cell state? A. It is reset to zero. B. It is approximately preserved across the timestep. C. It is replaced by the candidate `ĉ_t`. D. It oscillates. **(1)**

> ***B.*** `c_t = f_t ⊙ c_{t-1} + i_t ⊙ ĉ_t ≈ c_{t-1}`*. This is the "constant error carousel" that lets information propagate over long horizons.*

1.10 True/False: In RLHF, the KL-regularisation term penalises the *current policy* for diverging from the *reward model*. **(1)**

> *False. KL is taken against the* ***reference policy*** `π_ref` *(the frozen SFT model), not the reward model.*

1.11 Which instruction-tuning data method **inverts** the usual direction — taking unlabelled text as the response and asking the LM to invent a plausible instruction that would produce it? A. Self-Instruct B.

Evol-Instruct C. Orca D. Instruction Back-translation **(1)**

> ***D — Instruction Back-translation*** *(Humpback / Meta).*

1.12 True/False: In Bahdanau attention, multiple decoder positions may focus their attention on the *same* encoder position. **(1)**

> *True. Bahdanau attention is many-to-many — alignment is soft and unconstrained.*

1.13 Which agentic framework improves over trials via *verbal* reinforcement (self-reflection text stored in memory) *without* updating model weights? A. ReAct B. ReWOO C. Reflexion D. Self-Refine **(1)**

> ***C — Reflexion.*** *Self-Refine iterates within a single trial; Reflexion accumulates reflections across trials.*

1.14 In the Bradley–Terry reward model with score `r_ϕ(x,y)`, the probability that `y_1` is preferred to `y_2` is: A. `σ(r_ϕ(x, y_1)) / σ(r_ϕ(x, y_2))` B. `σ(r_ϕ(x, y_1) − r_ϕ(x, y_2))` C. `softmax over y_1, y_2 of r_ϕ` D. `r_ϕ(x,y_1) − r_ϕ(x,y_2)` **(1)**

> ***B.*** *Standard pairwise sigmoid formulation. (Option C is mathematically equivalent for two items but the canonical form is B.)*

1.15 True/False: The per-token training cost (FLOPs per token, ignoring batching) of a Transformer encoder is `O(T·d)` for fixed model size `d` and sequence length `T`. **(1)**

> *False. The attention layer is `O(T² d)` per layer per forward pass, so the per-token training cost is `O(T d + d²)`, growing linearly with `T`.*

1.16 Which CNN block computes a per-channel scaling vector via Global-Average-Pool → bottleneck-FC → sigmoid? A. Spatial attention B. SE (Squeeze-and-Excitation) C. CBAM D. Dropout **(1)**

> ***B — Squeeze-and-Excitation.*** *CBAM uses SE plus a separate spatial attention.*

1.17 True/False: In PPO-CLIP, when the advantage is positive and the policy ratio exceeds `1+ε`, the surrogate objective continues to receive gradient signal proportional to `r_t · Â_t`. **(1)**

> *False. In that region the surrogate becomes `(1+ε)·Â_t`, which is a constant w.r.t. θ — no gradient. That is precisely the clipping mechanism that prevents over-large updates.*

1.18 Mask R-CNN replaces RoIPool with **RoIAlign** primarily to: A. speed up training. B. avoid quantisation of spatial coordinates which harms pixel-accurate masks. C. allow variable batch sizes. D. enable multi-scale features. **(1)**

> ***B.*** *RoIPool snaps to integer coordinates twice (region → cell), introducing misalignment that is fine for boxes but ruinous for pixel masks.*

1.19 Which positional encoding scheme makes the dot product `q_i · k_j` depend only on the *relative* offset `i - j`? A. Sinusoidal absolute B. Learned absolute C. RoPE D. ALiBi **(1)**

> ***C — RoPE.*** *Rotations encode position so that the inner product collapses to a function of* `i-j`*. ALiBi is a different mechanism (additive linear bias).*

1.20 In the HuggingGPT pipeline, which stage maps each parsed sub-task to a concrete model on the Hugging Face Hub? A. Task Planning B. Model Selection C. Task Execution D. Response Generation **(1)**

> ***B — Model Selection.*** *Task Planning produces the sub-task DAG; Model Selection chooses which model handles each sub-task.*

Section 2 — CNN regularisation (5 marks)

2.1 Explain *dropout* as applied in CNNs, contrasting plain dropout with **SpatialDropout** / **DropBlock**, and state when each is preferred. **(2)**

> *Plain dropout zeroes out individual activations independently with probability* `p` *and scales the rest by* `1/(1-p)` *(inverted dropout). In CNN feature maps, spatially adjacent activations are highly correlated, so independent pixel dropout removes very little information — the network can simply infer the dropped value from neighbours.* ***SpatialDropout*** *drops entire feature-map channels, forcing reliance on different filter responses.* ***DropBlock*** *drops contiguous spatial regions, removing semantically meaningful patches. SpatialDropout is preferred for thin networks where channel co-adaptation is the main risk; DropBlock is preferred for large CNNs on natural images (e.g. ResNet on ImageNet).*

2.2 Write the BatchNorm transform and explain why it permits higher learning rates and acts as a regulariser. Also state what happens at inference. **(3)**

> *During training, for a mini-batch B and per-channel statistics:*
>
> ```
> μ_B = (1/m) Σ x_i      σ²_B = (1/m) Σ (x_i − μ_B)²
> x̂_i = (x_i − μ_B) / √(σ²_B + ε)
> y_i  = γ · x̂_i + β
> ```
>
> *Higher LR: by fixing activation statistics, BN linearises the optimisation landscape — gradients propagate to deeper layers without saturating; Lipschitz of the loss with respect to weights is bounded more tightly (Santurkar et al.). Regulariser: μ_B and σ^2_B are noisy estimates from each mini-batch, injecting stochasticity into activations — similar in spirit to dropout, though smaller in magnitude. Inference: replace* `μ_B, σ²_B` *with running EMA statistics accumulated during training.* `γ, β` *remain learnable params.*

Section 3 — Attention basics (5 marks)

3.1 Define attention in one sentence and give one NLP example and one CV example. **(1)**

> *Attention learns to weight a set of input elements by relevance to a query, producing the output as a weighted combination of those elements. NLP: in machine translation, when generating "chat" the model focuses attention on the English "cat". CV: in image captioning, when generating "ball" the model focuses on the pixel region containing the ball.*

3.2 Distinguish **spatial** from **channel/spectral** attention, with one application of each. **(2)**

> *Spatial attention assigns weights over H×W locations and is multiplied into the feature map to emphasise relevant regions — useful for fine-grained recognition, e.g. focusing on a small lesion in a medical image. Channel/Spectral attention (SE) assigns weights over the C channels and rescales them — useful in hyperspectral imagery, where different spectral bands carry different discriminative information (NIR vs RGB for vegetation).*

3.3 State the *scaled-dot-product self-attention* equation and explain the role of the `√d_k` scaling. **(2)**

> ```
> Attention(Q,K,V) = softmax(QK^T / √d_k) · V
> ```
>
> *For independent zero-mean unit-variance entries, the variance of an individual dot product `q·k` grows linearly with `d_k`. Without scaling, large dot products would push the softmax into very peaky regions, vanishing the gradients for non-maximum entries. Dividing by `√d_k` keeps the pre-softmax variance ≈ 1.*

Section 4 — Transformer details (6 marks)

4.1 Explain the motivation for **Multi-Head Attention** and write the equation that combines heads. **(2)**

> *Softmax forces a single attention head to be peaky, so it can only model one dominant relationship per query position. Multi-Head Attention projects Q, K, V `h` times into lower-dimensional subspaces and runs attention in parallel, letting different heads attend to different relations (e.g. subject–verb in one head, coreference in another). Outputs are concatenated and linearly mixed:*
>
> ```
> MHA(X) = Concat(head_1, …, head_h) · W_O,  head_i = Attn(X W_Q^i, X W_K^i, X W_V^i)
> ```

4.2 Explain causal masking in the decoder, give the mathematical form, and state precisely *what* property it preserves. **(2)**

> *During teacher-forced training of an auto-regressive decoder, the model sees the whole target sequence in parallel. To prevent token* `l` *from attending to any future token* `t > l`*, we set the attention logit* `e_{l,t} = −∞` *before the softmax — those weights become 0. This preserves the* ***auto-regressive property****: the prediction at position* `l` *depends only on positions* `≤ l`*, matching what is available at inference time.*

4.3 Sinusoidal vs learned absolute vs RoPE positional encoding — name one strength and one weakness of each. **(2)**

> *Sinusoidal: + extrapolates beyond training length in principle; + no params. − fixed; − absolute only, no nice relative behaviour without tricks. Learned absolute: + flexible, optimised for task. − cannot extrapolate beyond training length; − more params. RoPE: + dot products depend only on relative offset; + extrapolates moderately. − more compute per layer for the rotation; − some saturation at very long distances.*

Section 5 — Pre-training & fine-tuning (4 marks)

5.1 Compare BERT's **MLM** with GPT's **causal LM** objective. Why is MLM unsuitable for free-form generation? **(2)**

> *MLM corrupts ~15% of tokens with* `[MASK]` *and trains the model to predict them using* ***bidirectional*** *context. Causal LM predicts the next token using* ***only the left context****. MLM is unsuitable for generation because (a) the model never learns to predict a full* sequence left-to-right, only fill in gaps; (b) at inference, all positions to the right are unknown — the bidirectional context BERT relies on does not exist; (c) decoding strategies designed for AR LMs don't directly apply.**

5.2 A practitioner has a 7B base LM and wants to fine-tune for legal-document summarisation with a single 24 GB GPU. Recommend a PEFT method and justify in 2-3 lines. **(2)**

> *Use* ***LoRA*** *(or QLoRA if memory is tight). LoRA injects low-rank updates* `BA` *(typically* `r=8–16`*) into each attention projection, training ~0.1% of parameters. On a 24 GB GPU, this comfortably fits a 7B model with adapters; QLoRA + 4-bit quantisation gives further headroom. At inference,* `W ← W + BA` *can be merged so there is no latency overhead.*

Section 6 — Prompting (2 marks)

6.1 Explain why **Chain-of-Thought** prompting improves arithmetic / multi-step reasoning over zero-shot prompting, and state when CoT *fails to help*. **(2)**

> *CoT exposes intermediate reasoning steps in the few-shot demonstrations (or via "Let's think step by step." zero-shot). Because the model now spends output tokens on intermediate computations, it can use the extra "thinking budget" to decompose the problem before answering. Empirically, the benefit is large for problems requiring multi-step reasoning and emerges only at scale (>~60B). CoT fails to help when (a) the task is single-step (sentiment classification), (b) the model is too small to produce coherent reasoning, or (c) the chain itself is internally inconsistent and propagates errors (mitigated by self-consistency).*

Section 7 — Instruction tuning & alignment (4 marks)

7.1 Why is SFT alone insufficient for safety? Give a concrete failure example. **(2)**

> *SFT optimises the likelihood of `(prompt, response)` pairs but does not let humans express preferences* among candidate responses. A model SFTed on `(question, helpful-answer)` pairs will be fluent and on-topic but has no signal to prefer harmless over harmful continuations. Example: prompt "How can I dispose of a competitor's online reviews?" — an SFT model may produce a step-by-step guide because "be helpful" is the only optimisation pressure. RLHF can rank a refusal above the harmful guide using preference data, fixing this.*

7.2 Briefly describe the *Instruction Evolver* operations in Evol-Instruct. **(2)**

> *Evol-Instruct begins with a seed of simple instructions; the Evolver applies one of several operations to create a harder version: - Add constraints (e.g. word limit, format requirement); - Deepen (require more detail / depth); - Concretise (replace vague terms with specific instances); - Increase reasoning steps; - Complicate input (add noise / ambiguity to the input data). Each round produces a new generation; failures (off-topic, too easy, too hard) are filtered.*

Section 8 — RLHF math (5 marks)

8.1 In LLM-RL, define **state**, **action**, and **policy**. **(2)**

> *State `s_t`: the prompt `x` concatenated with all tokens generated so far `y_{<t}`. Action `a_t`: the next output token, drawn from vocabulary `V`. Policy `π_θ(a | s_t)`: the LLM's softmax distribution over the vocabulary, parameterised by θ.*

8.2 Write the **regularised reward-maximisation** objective used in RLHF and explain the role of each term. **(3)**

```
max_ϑ  E_{x~D, y~π_ϑ(·|x)} [ r_φ(x, y) ]  −  β · KL( π_ϑ(·|x) ‖ π_ref(·|x) )
```

or equivalently as a per-sample reward:

```
E_{π_ϑ(y|x)} [ r_φ(x,y) − β · log( π_ϑ(y|x) / π_ref(y|x) ) ]
```

- `r_φ(x,y)`*: scalar reward from the pre-trained reward model (Bradley–Terry).*
- `π_ref`*: frozen reference policy, usually the SFT model.*
- *KL term penalises drift from the reference policy — prevents reward hacking (gibberish that scores high) and preserves output diversity / fluency.*
- `β`*: regularisation strength; tuned on a held-out set.*

Section 9 — Algorithms (4 marks)

9.1 Explain *importance sampling* and how it justifies multi-epoch gradient updates in PPO. **(2)**

Importance sampling rewrites an expectation under one distribution in terms of samples from another:

```
E_{π_ϑ}[f(x)] = E_{π_old}[ (π_ϑ(x)/π_old(x)) · f(x) ].
```

In PPO we collect rollouts under `π_old` *(the policy at the start of the iteration), then take several gradient-update epochs while reweighting by the ratio* `r_t = π_ϑ/π_old`*. Without IS, each epoch's gradient would be biased after the first update; IS keeps it unbiased provided the ratios are bounded — which PPO ensures by clipping.*

9.2 Write the **PPO-CLIP** surrogate objective and explain how the clipping prevents catastrophic policy updates. **(2)**

```
L^{CLIP}(ϑ) = E_t [ min( r_t · Â_t,  clip(r_t, 1−ε, 1+ε) · Â_t ) ]
```

Case `Â_t > 0`*: if* `r_t > 1+ε`*, the surrogate is capped at* `(1+ε)·Â_t`*, so gradient ascent stops trying to push the probability of this good action higher. Case* `Â_t < 0`*: if* `r_t < 1−ε`*, the surrogate is capped at* `(1−ε)·Â_t`*, so gradient stops trying to push the probability lower. The* `min` *over clipped and unclipped objectives yields a **pessimistic lower bound** — updates outside the trust region get no signal, preventing destabilising swings.*

Section 10 — Retrieval & Agents (3 marks)

10.1 Differentiate **RAG-Sequence** from **RAG-Token** and explain which is more appropriate when the answer must be sourced from a single coherent document (e.g. legal QA). **(2)**

> *RAG-Sequence: retrieve top-K passages, generate the full output once per passage, then marginalise over passages — each candidate output is grounded in **one** document. RAG-Token: retrieve top-K and, at every output token, marginalise over passages — different tokens may cite different documents. For legal QA where mixing sources is problematic (laws and clauses can be contradictory across jurisdictions), **RAG-Sequence** is more appropriate: the final answer is traceable to a single document.*

10.2 ReWOO claims efficiency over ReAct. Explain in one sentence how. **(1)**

> *ReWOO produces an **entire reasoning plan with placeholders in a single LLM call**, then a separate worker executes tools in batch and a solver fuses results — avoiding ReAct's per-step LLM call that re-reads the full Thought/Action/Observation history.*

Section 11 — Design problem (6 marks)

11. You are asked to build a customer-support chatbot for an electronics retailer. The bot must (i) handle product questions using the company's catalogue, (ii) refuse harmful queries (e.g. instructions to circumvent device security), and (iii) hand off to a human if confidence is low. You have access to a 7B open-source LM, a small set of FAQ pairs (~5k), and a vector store of catalogue documents.

11.1 Propose a full training pipeline (data → model). Name the techniques used at each stage. **(3)**

> *Pipeline: 1. **SFT (Instruction tuning) on FAQ pairs.** Format as (instruction, response); train with cross-entropy. If 5k is too small for the 7B base, augment with **Self-Instruct** to synthesise additional electronics-domain instructions and **Evol-Instruct** to harden them. 2. **Reward modelling.** Collect ~10k human pairwise preferences on (helpful, harmless, on-topic) responses; train a Bradley–Terry reward model initialised from the SFT model + scalar head. 3. **RLHF (PPO + KL).** Maximise `E[r_φ(x,y)] − β · KL(π_θ ‖ π_SFT)` to align the SFT model with preferences. 4. **Use PEFT (LoRA)** at every fine-tuning stage so a single 24 GB GPU suffices. 5. **(Optional) Constitutional AI** to handle harmlessness at scale without expensive human labels — write principles, generate self-critique pairs, train a safety reward.*

11.2 Propose how the deployed system answers a question; specify retrieval, generation, and refusal logic. **(2)**

> 1. *On user query, **embed** the query and retrieve top-K (say K=5) passages from the catalogue vector store (DPR-style dense retrieval).*
> 2. *Run a lightweight **safety classifier / Constitutional check** on the query; if harmful, refuse with a templated message.*
> 3. *Else, condition the SFT+RLHF model on `(system prompt + retrieved passages + query)` via **FiD-style** encoding (separate encoders fused in decoder) so multiple catalogue pages can contribute.*
> 4. *Compute a **calibrated confidence** (e.g. softmax-temperature-scaled length-normalised log-prob, or reward-model score). If below threshold τ, return a handoff message rather than the generated answer.*

11.3 What is the main risk of using **RAG-Token** vs **RAG-Sequence** in this setting, and which would you choose? Justify in two lines. **(1)**

> *RAG-Token may blend incompatible facts from different product datasheets (e.g. mix specifications from two laptop SKUs).* ***RAG-Sequence*** *keeps each candidate answer grounded in a single document — chosen for traceability and to avoid hallucinated cross-product claims.*

Total: 50 marks

Appendix — Bonus original problems (extra practice)

These are graded harder than the model paper; attempt after the mock.

B1. Given a Transformer encoder with `d_model = 768`, `h = 12` heads, `L = 12` layers and FFN expansion `4×`, compute the approximate parameter count of one encoder layer (ignore biases and LN). *Answer:* `MHA params = 4 · d² = 4 · 768² ≈ 2.36M` (Q, K, V, O projections). `FFN params = 2 · 4 d · d = 8 d² ≈ 4.72M`. Total per layer ≈ **7.08 M**.

B2. A reward model gives `r(x, y_w) = 2.5` and `r(x, y_l) = 1.0`. Under Bradley–Terry, what is `P(y_w ≻ y_l | x)`? *Answer:* `σ(2.5 − 1.0) = σ(1.5) ≈ 0.818`.

B3. Show that for the regularised RLHF objective `E[r] − β KL(π_θ ‖ π_ref)`, the closed-form optimal policy (over an unconstrained function class) is `π*(y|x) ∝ π_ref(y|x) · exp(r(x,y)/β)`. *Sketch:* form Lagrangian, take functional derivative w.r.t. `π_θ`, set to zero. This is the basis of **DPO**.

B4. A bigram LM is trained on a corpus where `c("the", "cat") = 3`, `c("the") = 1000`, `V = 50000`. Compute MLE and add-1 estimates of `P("cat" | "the")`. *Answer:* MLE = `3/1000 = 0.003`. Add-1 = `(3+1)/(1000+50000) ≈ 7.84 × 10⁻⁵`.

B5. In Bahdanau attention, the alignment score uses the **previous** decoder hidden state `s_{t-1}` rather than the current `s_t`. Why is this convenient, and how does Luong's choice differ? *Answer:* Bahdanau's score is computable *before* `s_t` is known, letting the same `c_t` feed into the recurrence that produces `s_t`. Luong's "global" attention computes `s_t` first using the previous context, then re-attends — slightly different gradient flow but empirically similar quality.

B6. Explain why FiD (Fusion-in-Decoder) scales better than concatenation-then-encode for K retrieved passages. *Answer:* Concatenation gives one sequence of length `K · L`, with attention cost `O((KL)²)`. FiD encodes each of the K passages independently in parallel (`O(K · L²)`) and only fuses in the decoder's cross-attention — much cheaper for large K.

B7. Give one situation where **Reflexion** outperforms **Self-Refine** despite using the same base model. *Answer:* Tasks with stochastic environments (e.g. a game with hidden state) — Self-Refine's single-trial loop cannot accumulate evidence about the environment, whereas Reflexion's cross-trial memory does.

B8. In PPO, what failure occurs if `β` (KL coefficient) is set to zero throughout RLHF training? *Answer:* The policy drifts arbitrarily far from `π_ref` to maximise reward, producing **reward-hacked** outputs (high reward-model score, low actual quality), and the language modelling distribution can collapse (modes drop out, fluency degrades).

— end of mock paper —